



IMAGE PROCESSING

Chapter 6: morphological IP

Dr. Cao Thi Luyen

Email: luyenct@utc.edu.vn

Tel: 0912403345

Application

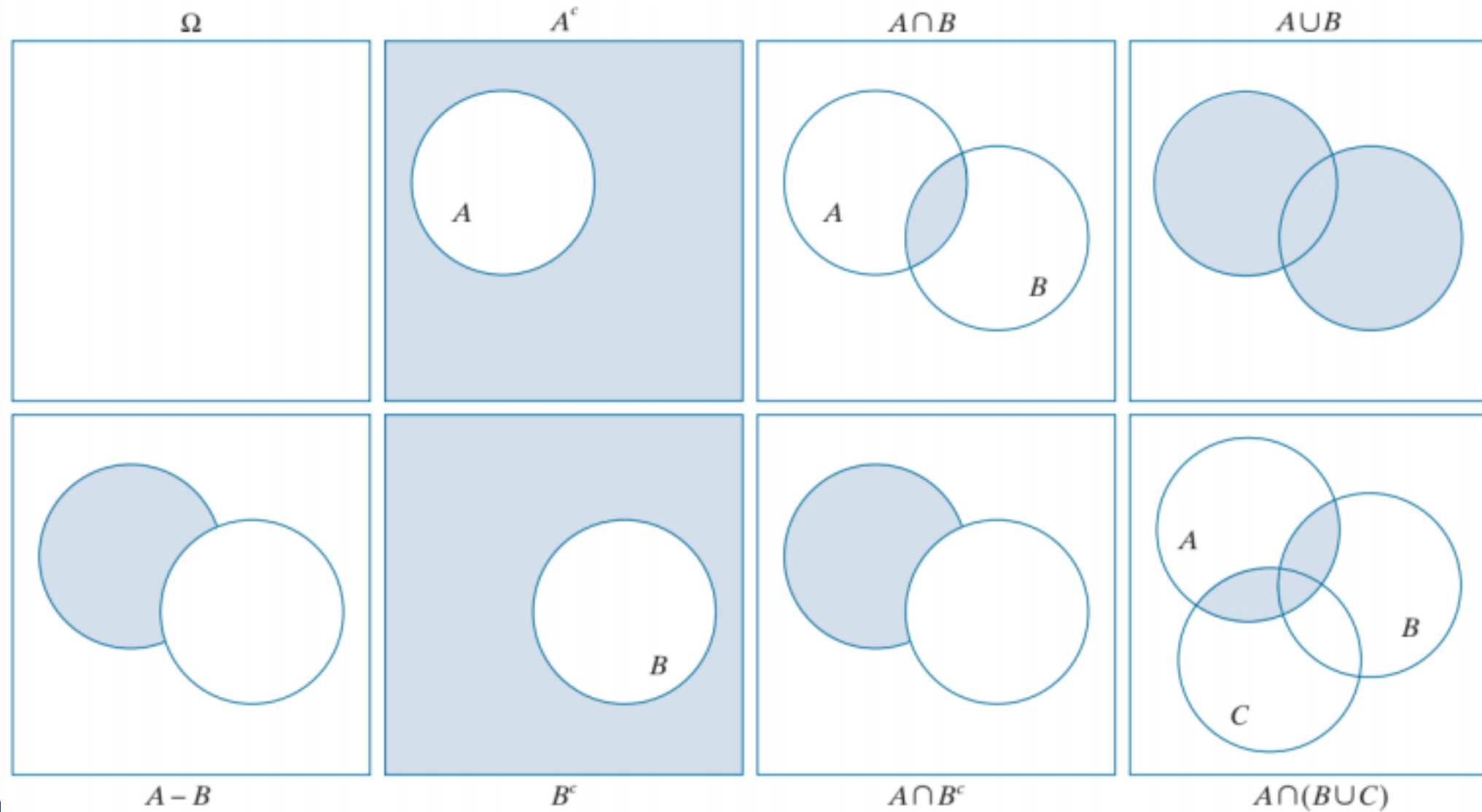


Image after segmentation

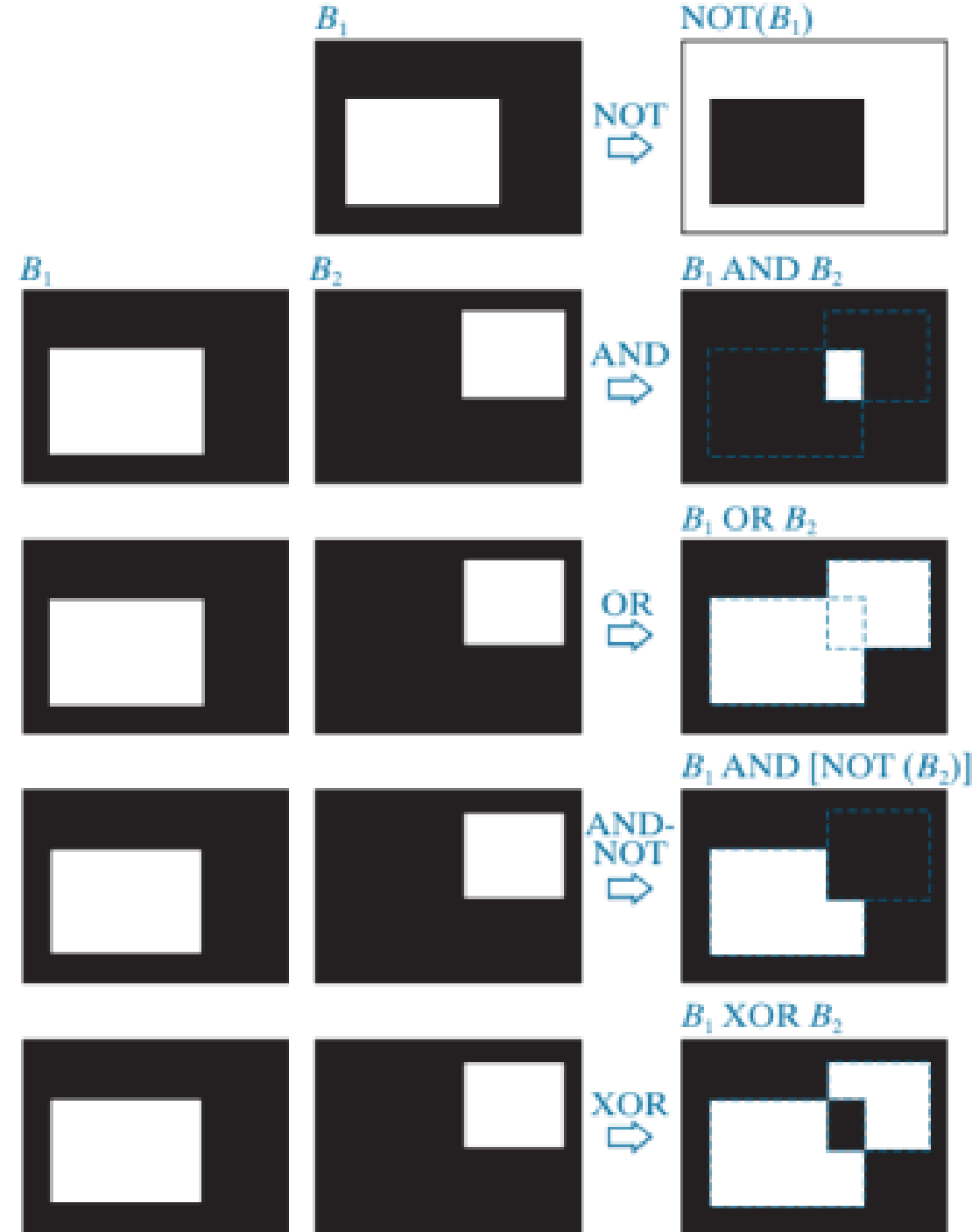


Image after segmentation and
morphological processing

Set theory

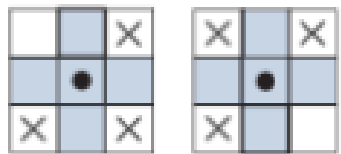
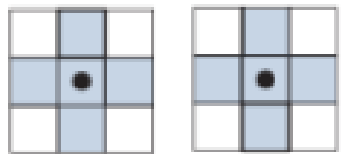
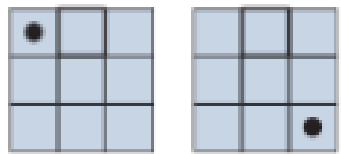


Logic operators



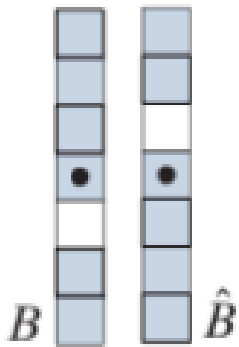
Reflection Of Structural Element

- **Structural element and its reflection through the center (.) is the center).** Reflection is a 180-degree rotation of the SE around the center.



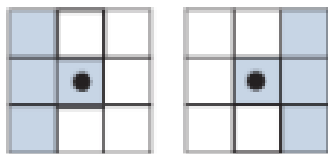
B

$\hat{B}$



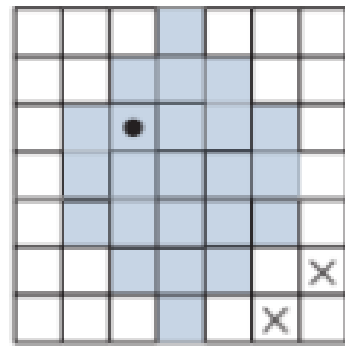
B

$\hat{B}$

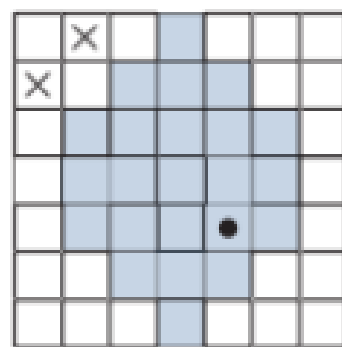


B

$\hat{B}$

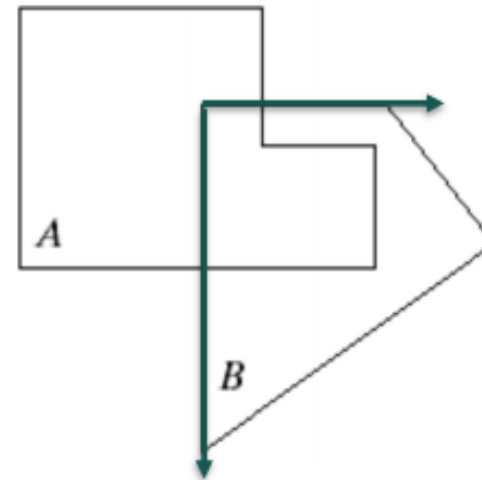
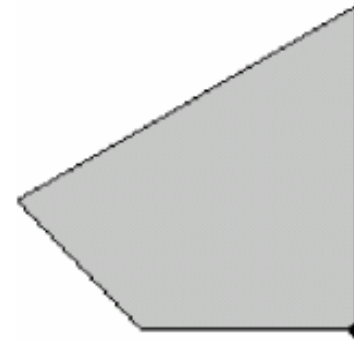


B



$\hat{B}$

$$\hat{B} = \{w | w = -b, b \in B\}$$



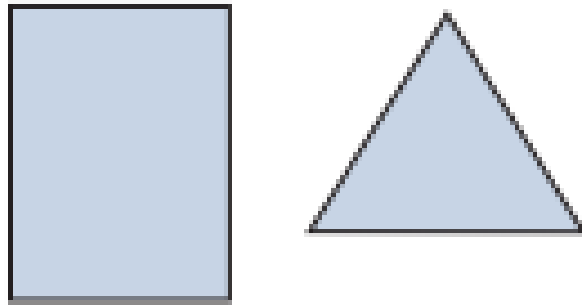
Structuring element

- **Structural Elements (SEs)** are matrices with structures created using two gray levels, 0 and 1.
- **Typically**, SEs are symmetric in shape with the centroid at the center.
- **Sometimes**, SEs contain “don't care” elements, denoted by $\times$.

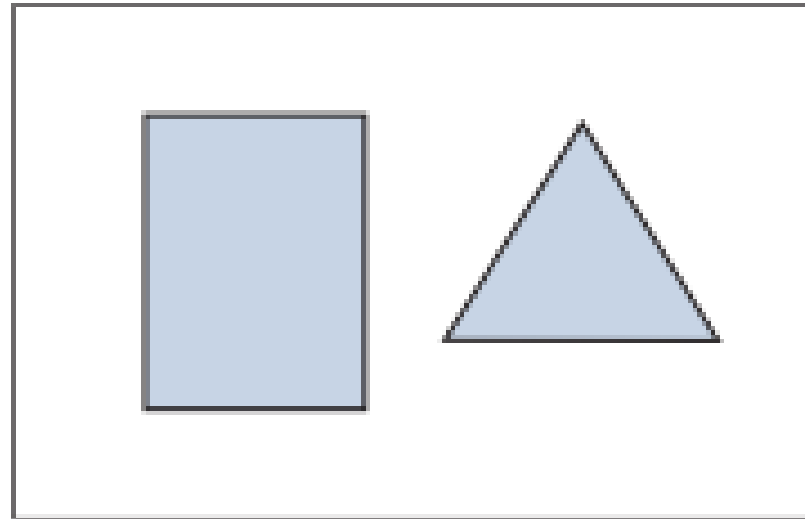
0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

0	1	0
1	1	1
0	1	0

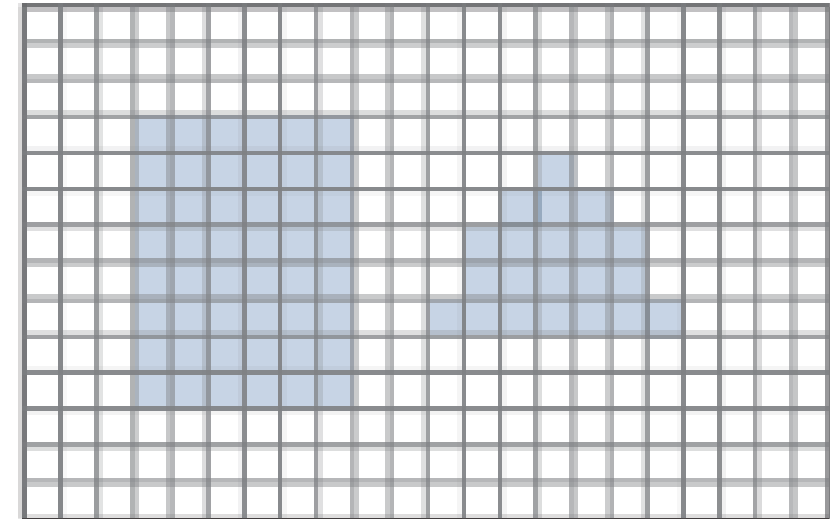
1	1	1
1	1	1
1	1	1



Objects represented
as sets



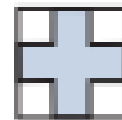
Objects represented as
a graphical image



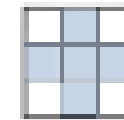
Digital image



Structuring element
represented as a set

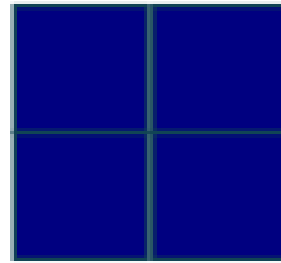
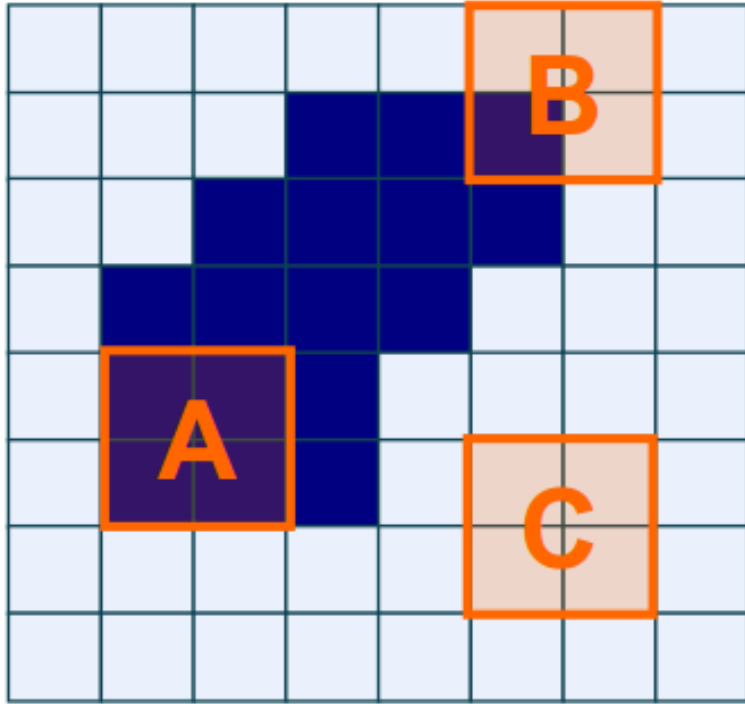


Structuring element
represented as a graphical image



Digital
structuring element

Hit, Fit, Miss



structural element

Fit: Fit occurs when all the pixels of the SE match the pixels of the image.

Hit: Hit occurs when any pixel of the SE matches a pixel of the image.

Miss: otherwise

0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0	0	0
0	0	1	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0

B

C

A

1	1	1
1	1	1
1	1	1

Structuring
Element 1

0	1	0
1	1	1
0	1	0

Structuring
Element 2

Basic Morphological Operations

- **Morphological image processing** is fundamentally similar to spatial filtering.
- The SE is moved across all pixels of the original image to create a new image.
- The value of the new pixels depends on the morphological operation.
- The two basic morphological operations are: erosion (shrink, reduce) and dilation (grow, expand).



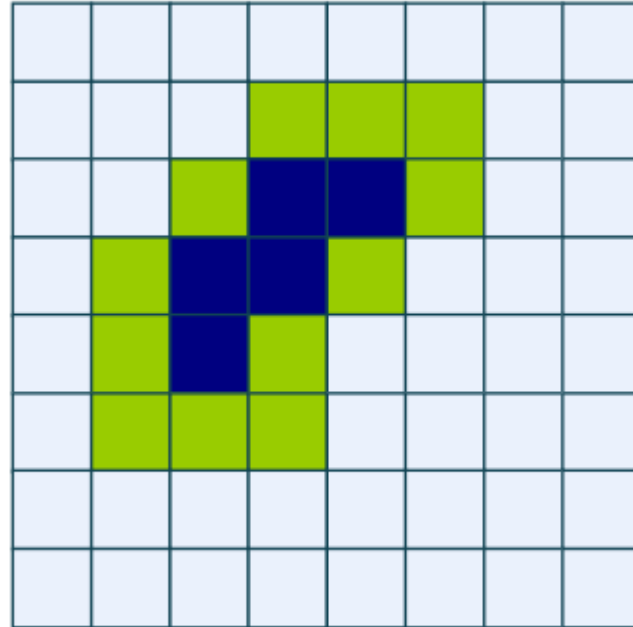
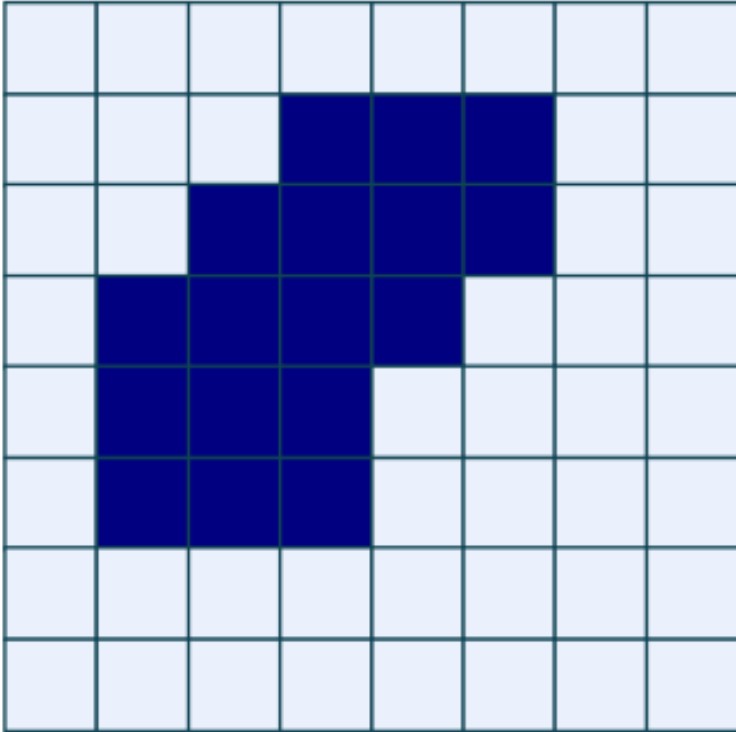
Erosion

- erosion equivalently as: $A \ominus B = \{z \mid (B)_z \cap A^c = \emptyset\}$

Assume SE B is at position (x,y). The new pixel value after performing the operation is as follows:

$$g(x, y) = \begin{cases} 1 & B \text{ fit } A \\ 0 & \text{otherwise} \end{cases}$$

Example



SE

Example



Original image



Erosion by 3*3
square structuring
element



Erosion by 5*5
square structuring
element

Application

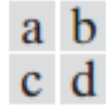


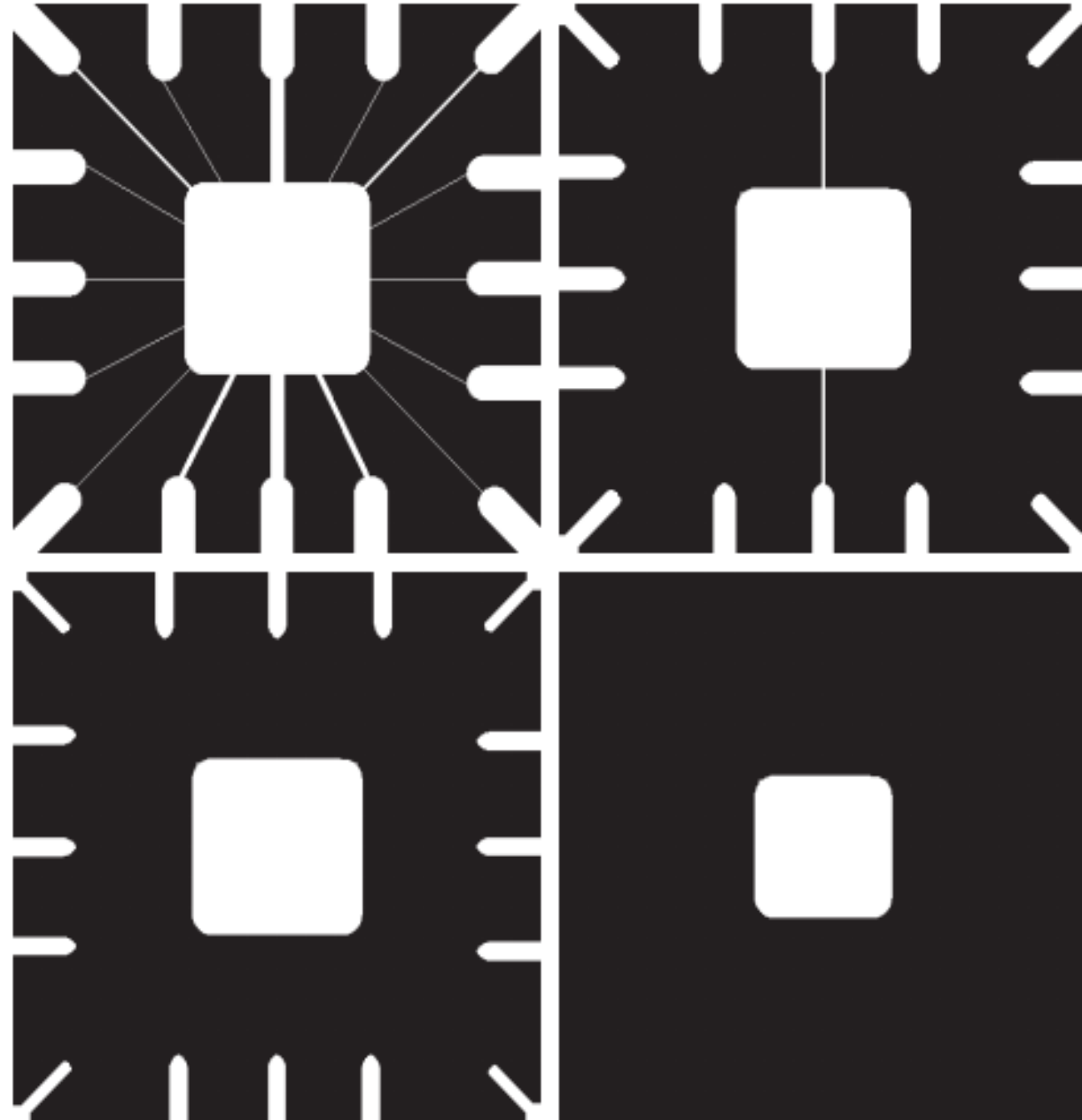
FIGURE 9.5

Using erosion to remove image components.

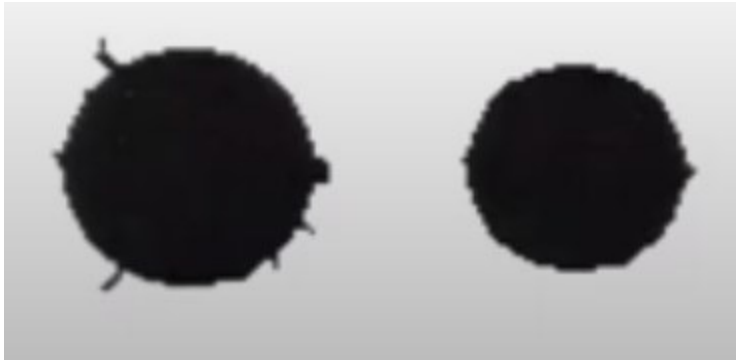
(a) A 486×486 binary image of a wire-bond mask in which foreground pixels are shown in white.

(b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 elements, respectively, all valued 1.

- Reduce the size of the object.
- Remove irrelevant details, trim the excess parts.
- Separate adjacent objects.



Application of Erosion



- Reduce the size of the object.
- Remove irrelevant details, trim the excess parts.
- Separate adjacent objects.

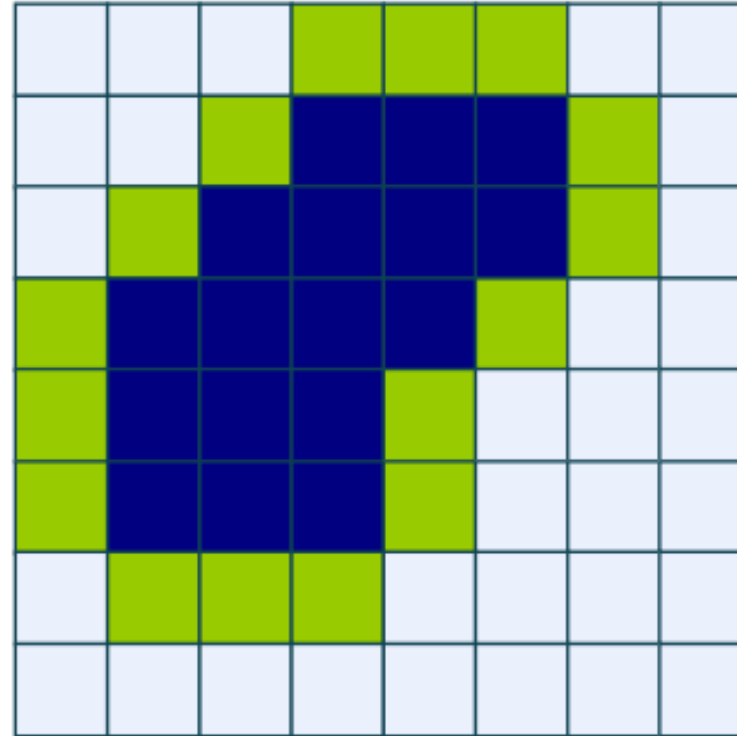
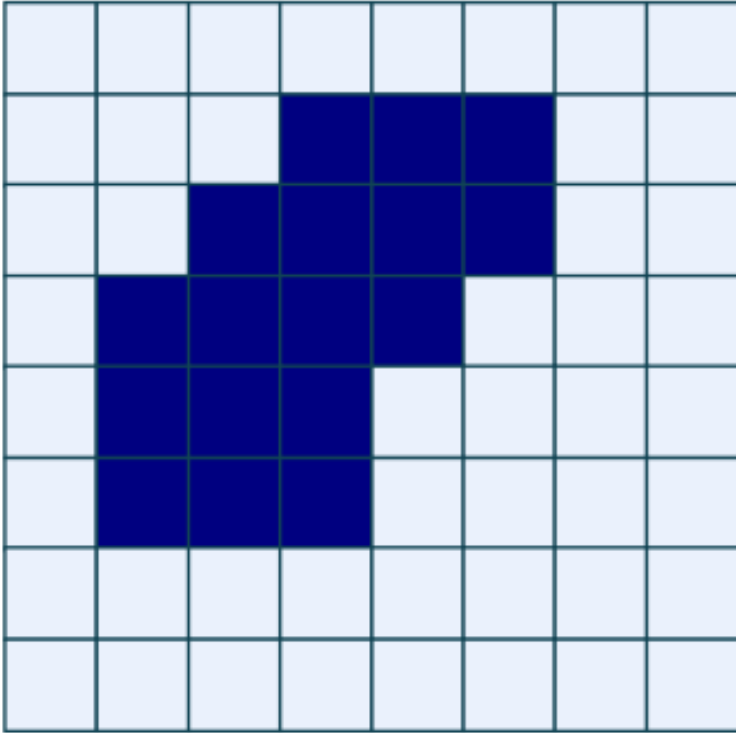
Dilation

$$A \oplus B = \left\{ z \mid (\hat{B})_z \cap A \neq \emptyset \right\}$$

Assume SE B is at position (x,y). The new pixel value after performing the operation is as follows:

$$g(x, y) = \begin{cases} 1 & \text{if } s \text{ hits } f \\ 0 & \text{otherwise} \end{cases}$$

Example





Original image

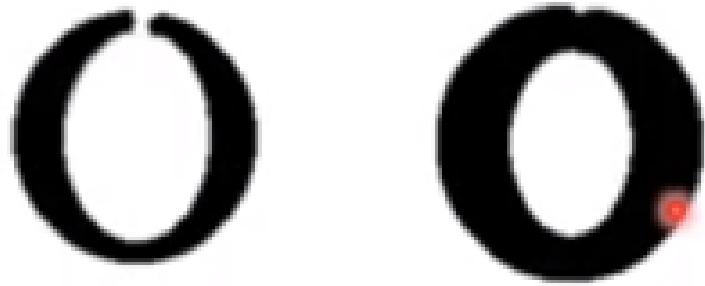


Dilation by 3×3
square structuring
element



Dilation by 5×5
square structuring
element

Application of Dilation



- Increase the size of the object.
- Fill in the missing parts of the object.
- Fix broken objects.

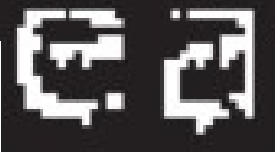


a c
b

FIGURE 9.7

- (a) Low-resolution text showing broken characters (see magnified view).
- (b) Structuring element.
- (c) Dilation of (a) by (b). Broken segments were joined.

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

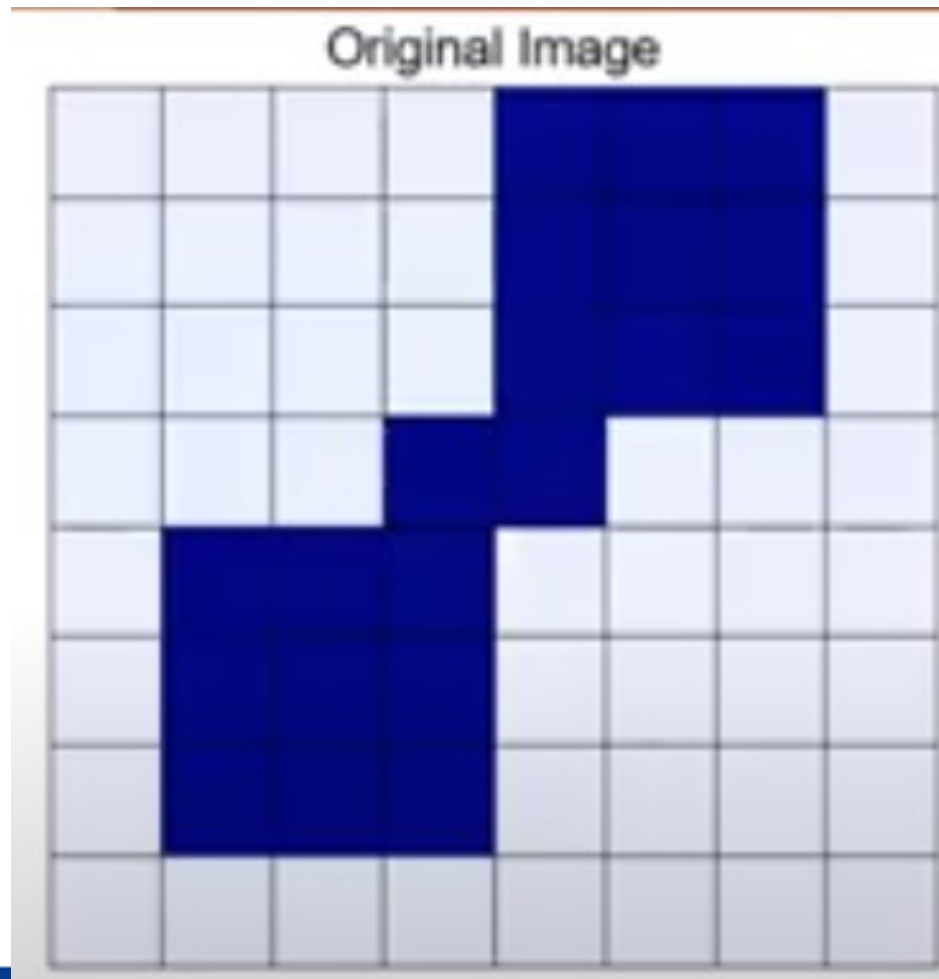


1	1	1
1	1	1
1	1	1

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



Practic with Erosion and dilation



Erosion and dilation Practice

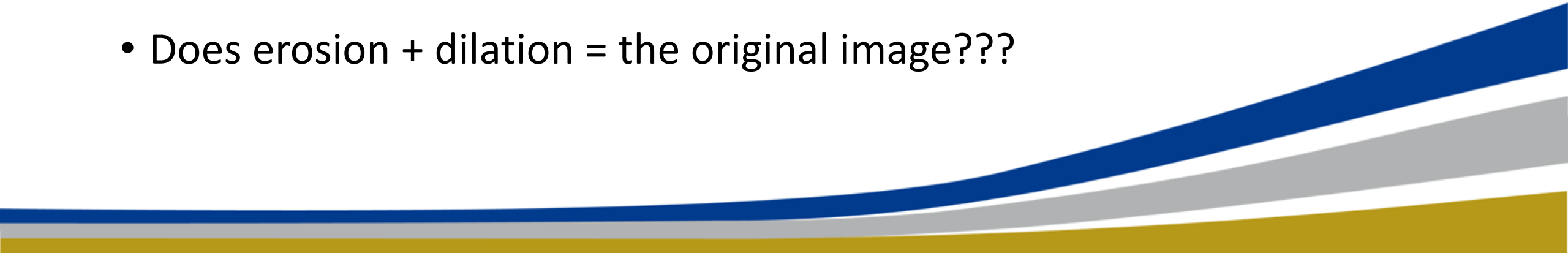
$$X = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Combining erosion and dilation

- **Erosion:** Shrinking the object.
- **Dilation:** Expanding the object with a structural element.
- **Desire:**
 - Remove structures or fill gaps.
 - Do not alter the remaining parts.
- **Solution:**
 - Combine erosion and dilation.
- Does erosion + dilation = the original image???

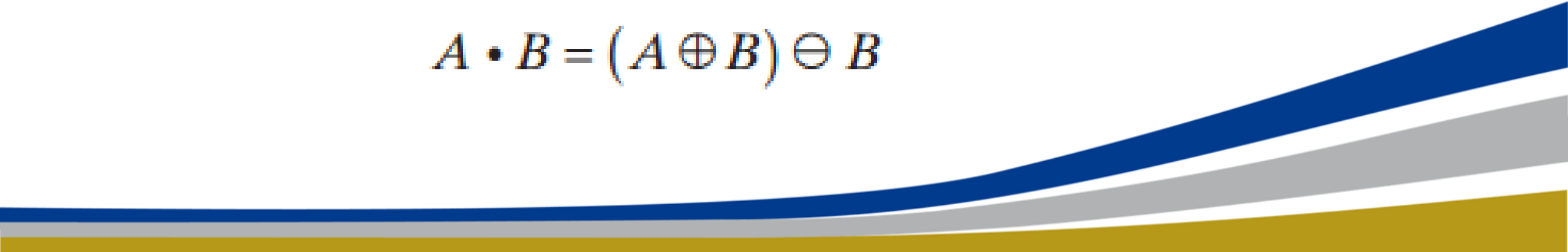


Closing and opening

- The *opening* of set A by structuring element B

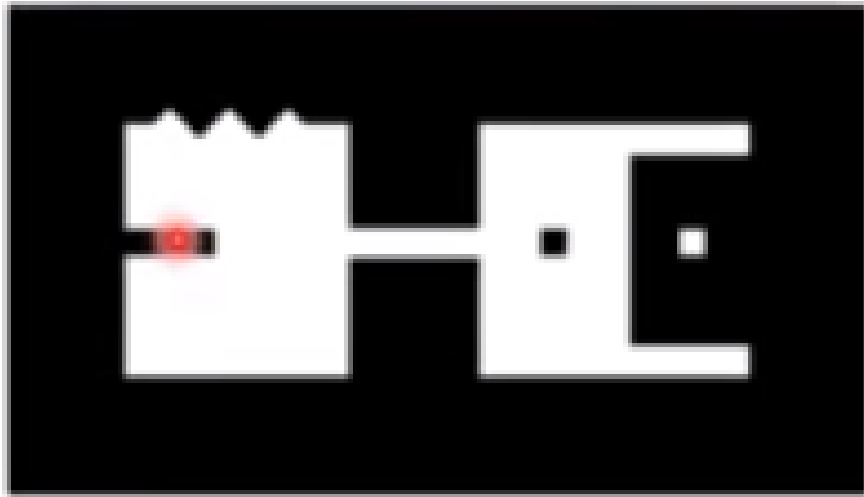
$$A \circ B = (A \ominus B) \oplus B$$

- the *closing* of set A by structuring element B

$$A \bullet B = (A \oplus B) \ominus B$$


Closing





Original Image



Image After Closing

Opening

a	b
c	d

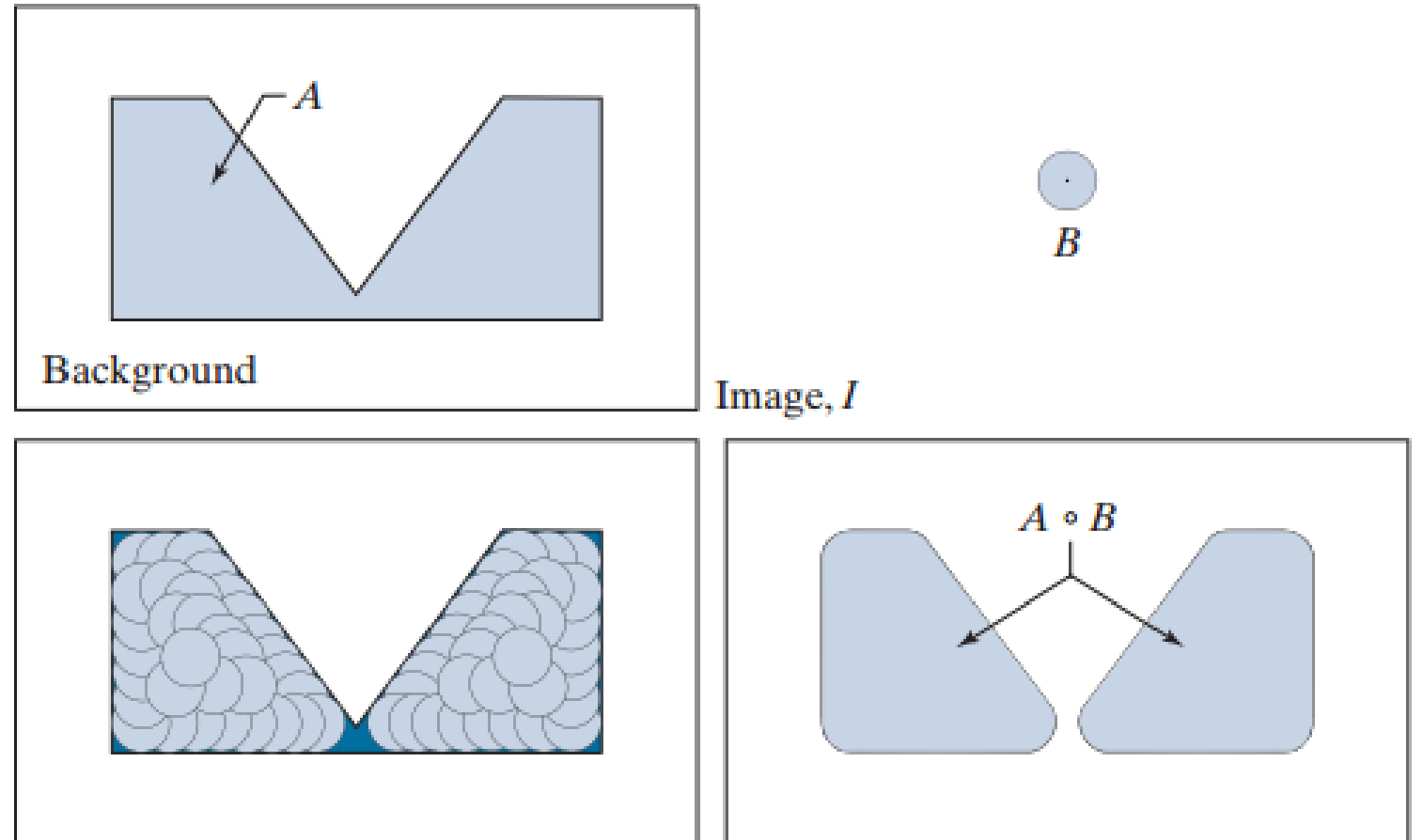
FIGURE 9.8

(a) Image I , composed of set (object) A and background.

(b) Structuring element, B .

(c) Translations of B while being contained in A . (A is shown dark for clarity.)

(d) Opening of A by B .



a	b
c	d

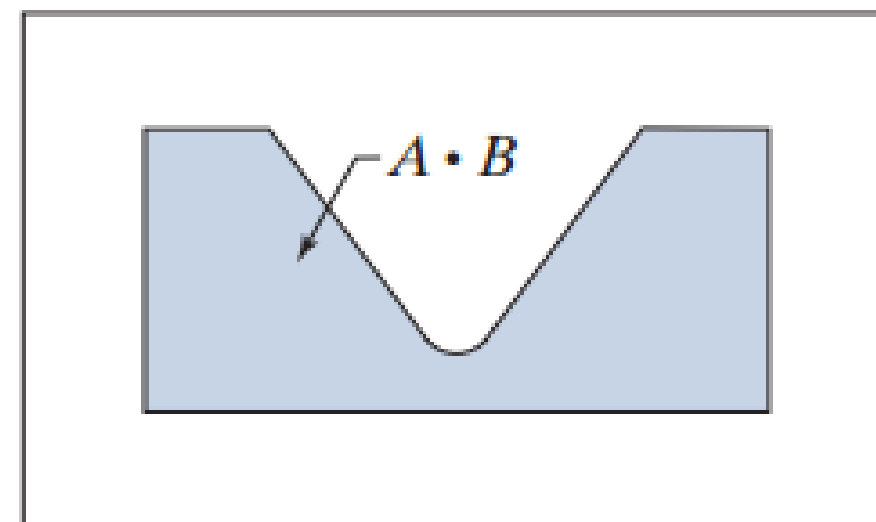
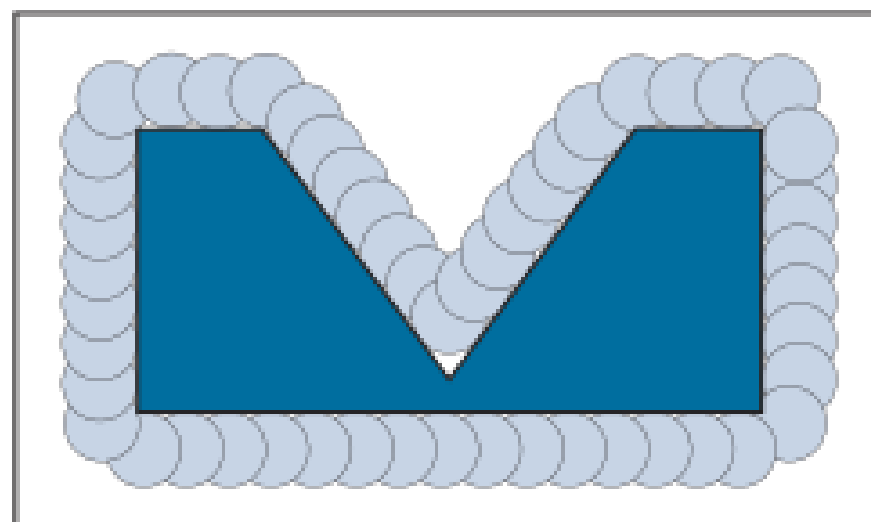
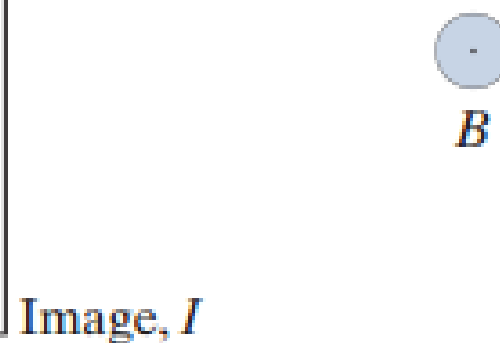
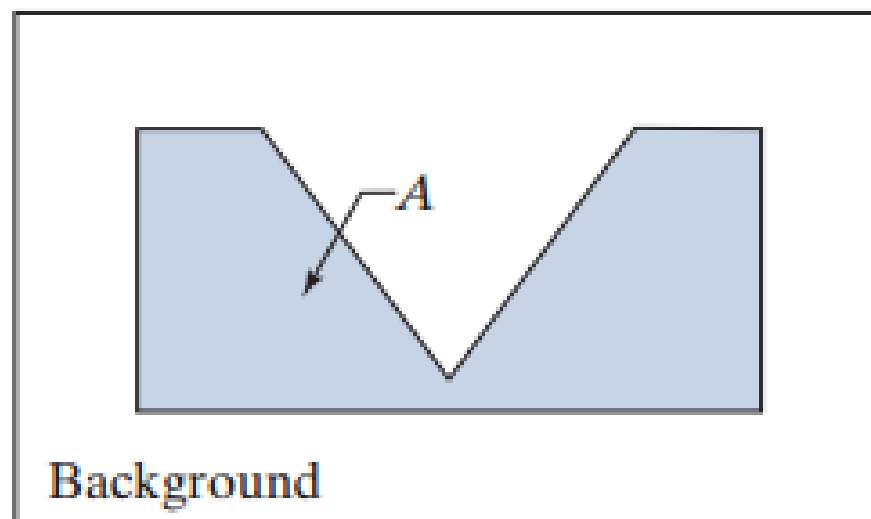
FIGURE 9.9

(a) Image I , composed of set (object) A , and background.

(b) Structuring element B .

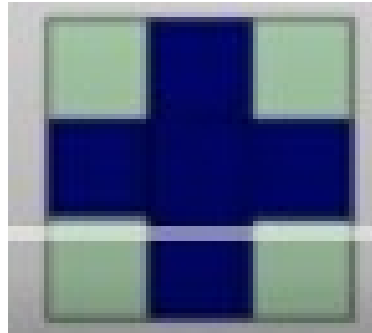
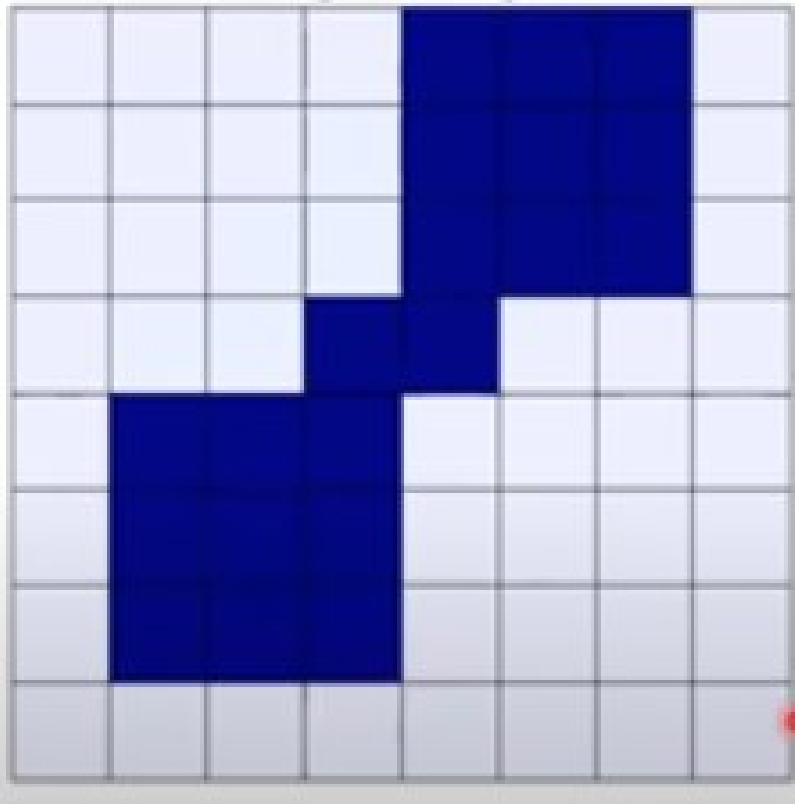
(c) Translations of B such that B does not overlap any part of A . (A is shown dark for clarity.)

(d) Closing of A by B .



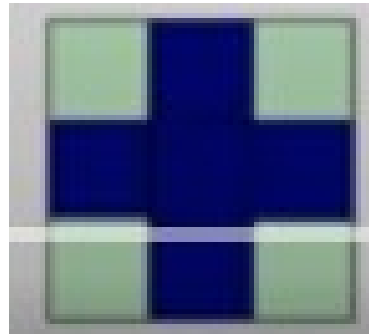
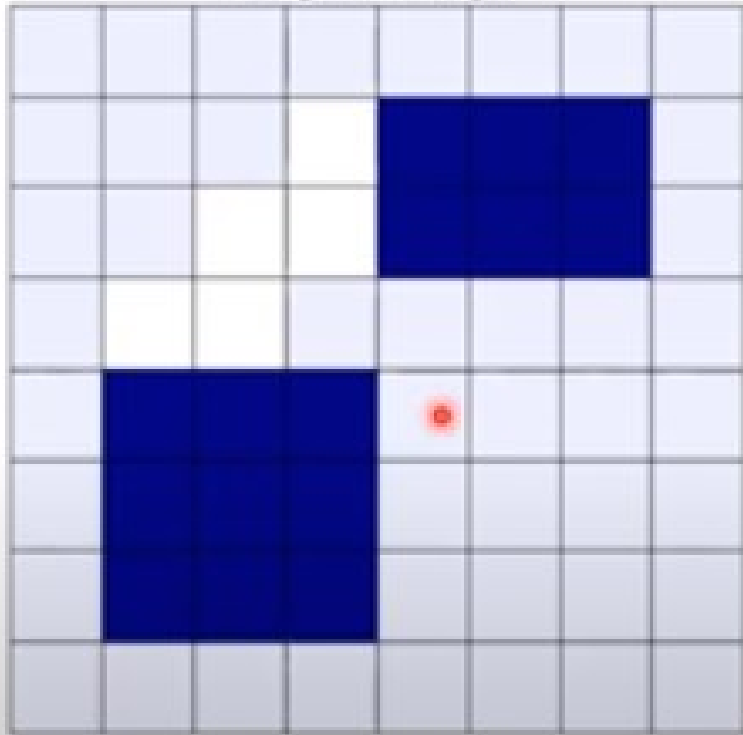
Opening?

Original Image

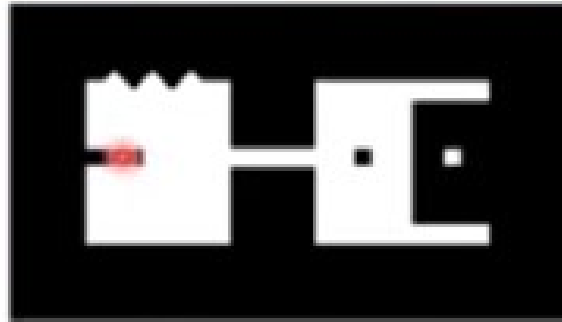


closing

Original Image



Example



Original Image

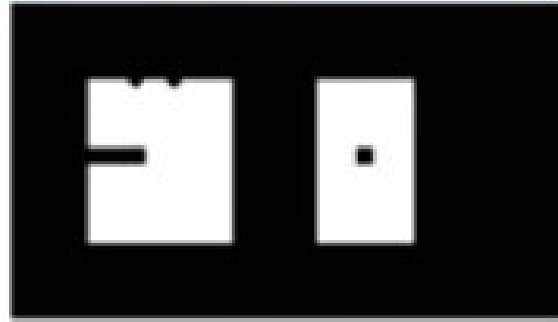
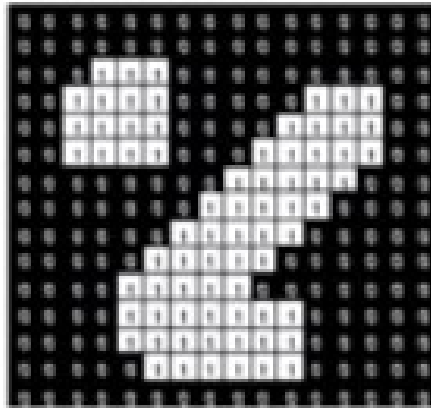
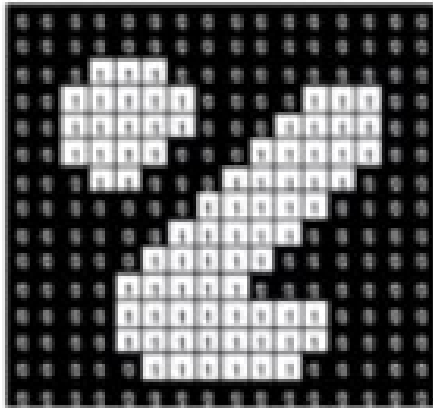


Image After Opening



Hit-or-miss transform (HMT)

HMT is a basic tool for object detection.

- I is a binary image consisting of foreground pixels (A) and corresponding background.
- HMT uses two structuring elements: B_1 to detect shapes in the foreground and B_2 to detect shapes in the background.

$$\begin{aligned} I \circledast B_{1,2} &= \{z | (B_1)_z \subseteq A \text{ and } (B_2)_z \subseteq A^c\} \\ &= (A \ominus B_1) \cap (A^c \ominus B_2) \end{aligned}$$

Boundary extraction

$$\beta(A) = A - (A \ominus B)$$



Hole filling

$$X_k = (X_{k-1} \oplus B) \cap I^c$$

$$k = 1, 2, 3, \dots$$

ring the found pixels



$$X_0 = p$$

$$X_k = (X_{k-1} \oplus B) \cap A^c, k = 1, 2, 3, \dots$$

form dilation

until

$$X_k = X_{k-1}$$

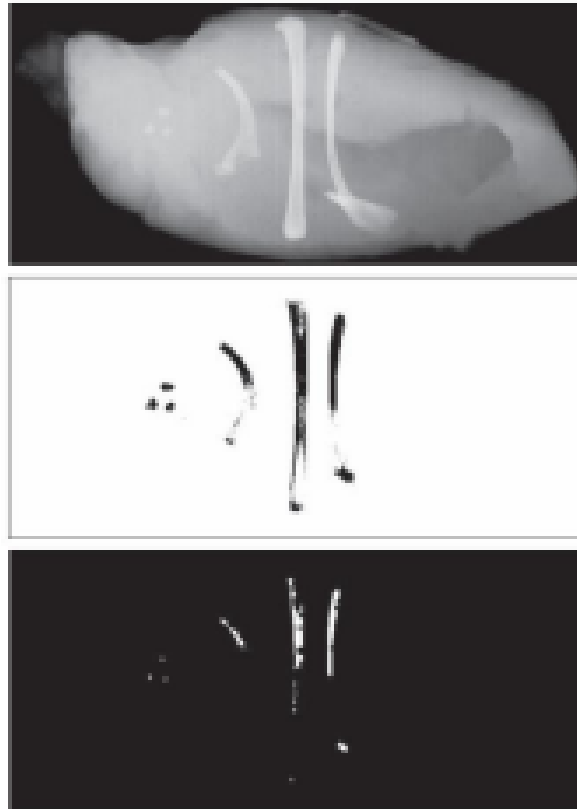
condition



Connected components

$$X_k = (X_{k-1} \oplus B) \cap A \quad k = 1, 2, 3, \dots$$

Start from a point within the connected component and perform dilation multiple times



Skeletons

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = (A \ominus kB) \\ = (A \ominus kB) \circ B$$

Reconstruction of A :

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

Thinning, Thickening

Thinning

$$\begin{aligned} A \otimes B &= A - (A \circledast B) \\ &= A \cap (A \circledast B)^c \end{aligned}$$

$$A \otimes \{B\} =$$

$$\left(\left(\dots \left((A \otimes B^1) \otimes B^2 \right) \dots \right) \otimes B^n \right)$$

$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$$

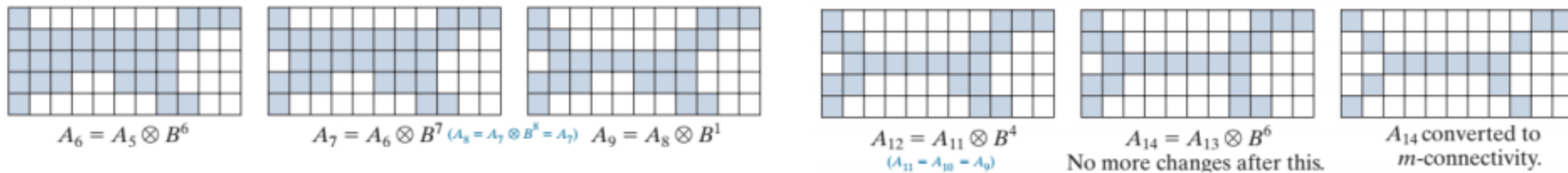
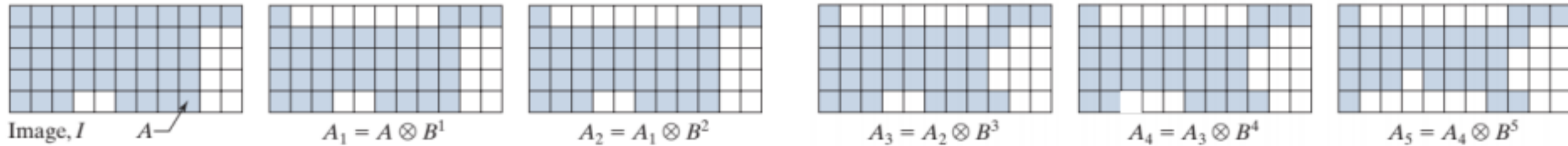
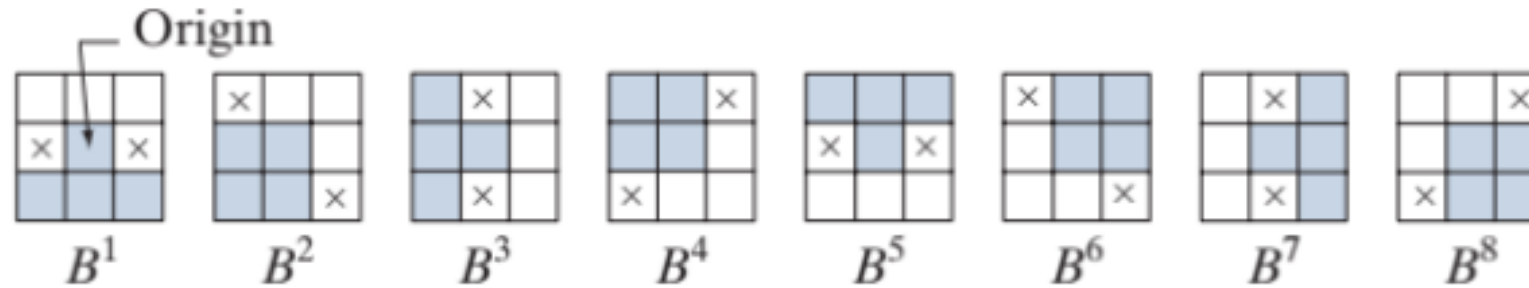
Thickening

$$A \odot B = A \cup (A \circledast B)$$

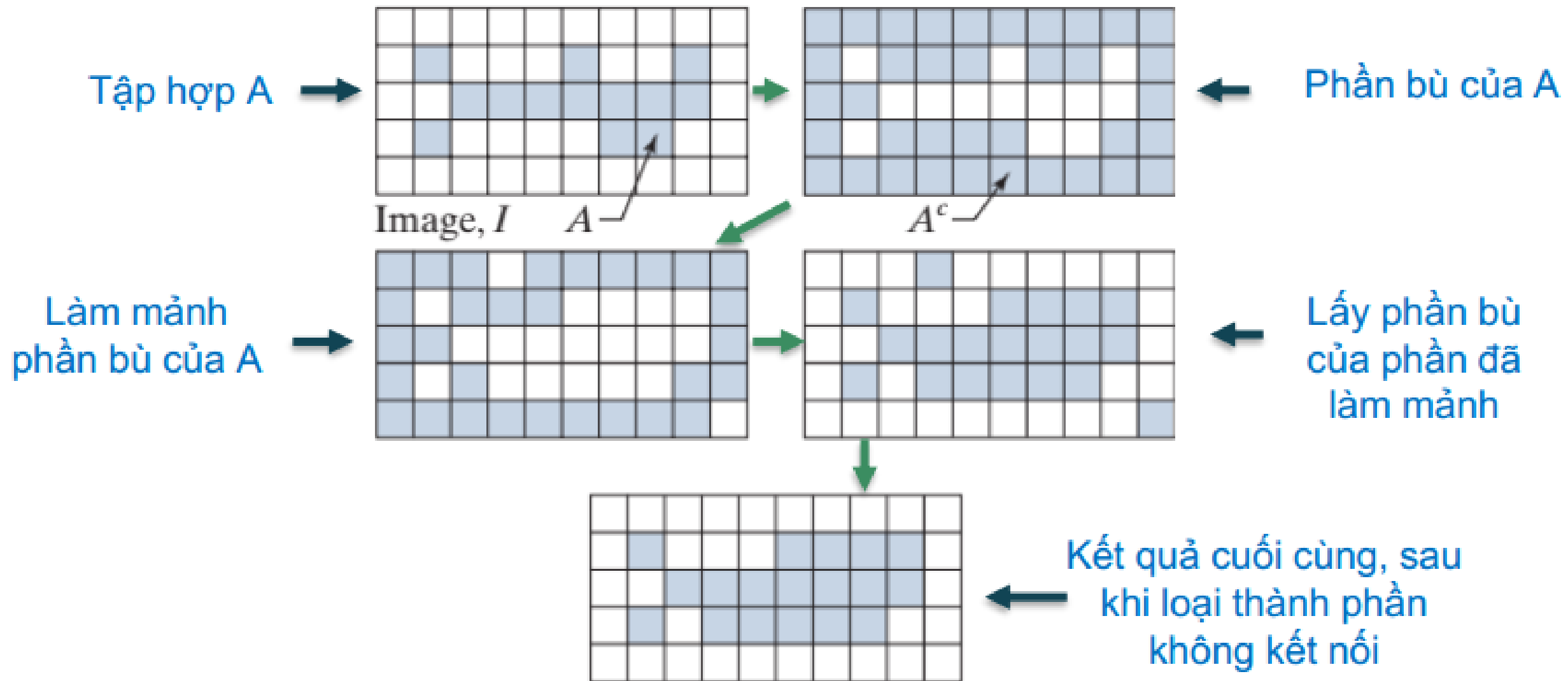
$$A \odot \{B\} =$$

$$\left(\left(\dots \left((A \odot B^1) \odot B^2 \right) \dots \right) \odot B^n \right)$$

Thinning example



Thickening example



Opening and closing Practice

$$X = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$